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Incremental capacity analysis based adaptive capacity estimation for lithium-ion battery considering charging condition☆

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HIGHLIGHTS

- A smoothing method for IC curves based on Kalman filter is proposed.
- Battery standard and non-standard charging experiments are designed and conducted.
- The influence of battery aging and charging condition on IC curves is revealed.
- The adaptive ICA for capacity estimation takes the charging initial SOC into account.

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ABSTRACT

Incremental capacity analysis (ICA) has been widely employed to investigate the degradation mechanism and perform the capacity estimation of lithium-ion batteries. However, the traditional capacity estimation based on ICA is limited by the computational efficiency and charging condition. In this paper, a rapid acquisition method of the incremental capacity (IC) curve is established, then an adaptive capacity estimation framework based on ICA considering the charging condition is proposed. Aiming at improving the computational efficiency, the Kalman filter is employed to acquire the smooth IC curves expeditiously, which requires small data length to handle the IC value. A considerable number of battery standard and non-standard charging experiments are designed and conducted. The influence of battery aging status, charging initial state of charge (SOC), and temperature on IC curves is revealed. Three features of the IC curve during standard charging are selected to investigate the relationship between battery capacity and the height of features. Furthermore, an adaptive correction method for capacity estimation considering the charging initial SOC is established, and the validation results show the effectiveness of the proposed correction method, which provides high accuracy and robustness.

1. Introduction

Under the trend of developing clean energy technologies and realizing automobile electrification, the lithium-ion battery has been deemed as the research focus due to its superior performance and environment friendliness [1–3]. As an essential parameter to characterize the battery performance, battery capacity is the preliminary information to estimate the battery state of charge (SOC) and state of health (SOH), as well as a reliable indicator of battery remaining useful life [4]. Therefore, it is essential to acquire the battery capacity accurately during the whole life cycle of the lithium-ion battery. Nevertheless, the nonlinear degradation of battery capacity challenges the battery management and capacity estimation method.

1.1. Literature review

The direct acquisition of battery capacity can be achieved by the discharge experiments. However, it is time-consuming and not available for vehicle applications due to strict operating conditions. Recently, several indirect methods of battery capacity acquisition/estimation have been proposed.

The model-based capacity estimation method has attracted many researchers' interest. Bartlett et al. [5] proposed a reduced-order electrochemical model and dual-nonlinear observers to estimate the SOC

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Nomenclature

CC	constant current
CC-CV	constant current-constant voltage
CLS	central least squares
DV	differential voltage
ETI	equal time interval
EVI	equal voltage interval
IC	incremental capacity
ICA	incremental capacity analysis
KF	Kalman filter
SOC	state of charge
SOH	state of health
Q_c	battery charging capacity
V_c	battery terminal voltage
i_c	battery charging current
Δt	the time interval for the ETI method
l_{KF}	the data length for the KF method
l_{CLS}	the data length for the CLS method
w_k	process noise
v_k	measurement noise
k	sample time
Q	process noise covariance
R	measurement noise covariance
x_k	the system state at the moment k
u_k	the system input at the moment k
y_k	the system output at the moment k
A_k	system matrix
B_k	input matrix
C_k	output matrix
D_k	feedthrough matrix
K_k	Kalman gain
$\hat{x}_k$	the estimated value of x_k
$\hat{x}_k^-$	*a priori* predicted value of the system state
$\hat{x}_k{}^+$	*a posteriori* updated value of the system state
P_k^-	*a priori* predicted value of the error covariance
P_k^+	*a posteriori* updated value of the error covariance
C_{F_i}	the estimated capacity according to the *i*th feature
H_{F_i}	the identified height of the *i*th feature
α_i	the slope for the *i*th feature to estimate capacity
β_i	the intercept for the *i*th feature to estimate capacity

and capacity for the battery with a composite electrode, and the battery aging mainly only considered the loss of cyclable lithium. Couto et al. [6] estimated the battery capacity with a single-particle model, and the capacity fade was indicated by solid surface concentration and diffusion time constant. For the traditional equivalent circuit model based capacity estimation, usually, a two-step estimation approach is employed. First, the battery SOC will be estimated through the battery open circuit voltage (OCV) [7] or some adaptive filters [8]. Then, the battery capacity will be obtained by reconstructing the SOC definition. Some adaptive filters, such as extended Kalman filter (KF) [9], total least squares algorithm [10,11] and so forth, are competent in battery capacity estimation and enhance accuracy. With the development of artificial intelligence, some intelligent data-driven algorithms have also attracted researchers' attention. Both the machine learning [12,13] and deep learning [14,15] have proved their effectiveness in battery capacity or SOH estimation. A great deal of experimental data is essential for the data-driven method; moreover, how to guarantee the quality of experimental data and accelerate the training speed is fundamental.

Another practical approach to estimate battery capacity is the incremental capacity analysis (ICA), which was initially proposed to investigate the battery degradation mechanism [16,17]. By calculating the derivative of the charging/discharging capacity to battery voltage, the ICA transferred the voltage plateaus caused by the electrode phase transition into intuitive and identifiable peaks on the incremental capacity (IC) curve. The reduction, shift, or disappearing of the IC peaks will be employed to reveal the degradation modes quantitatively. ICA is considered as one of the most advanced and non-destructive technologies to identify the battery fading mechanisms [18], which has been employed for the batteries with $Li(NiCoAl)O_2$ (NCA) [16,19], $LiFePO_4$ (LFP) [17,18], $Li(NiCoMn)O_2$ (NCM) [20,21], and NCM + $LiMn_2O_4$ (LMO) [22,23] cathode. A charging/discharging current with C/25 is usually adopted to detect the aging mechanism because the battery can be considered in an equilibrium state approximately with such a low current [17,19]. Although a large current will lead to the slight deformation of the IC curves due to the polarization effect, the ICA is still a promising method to estimate battery capacity [20].

The foundation of the capacity estimation based on ICA is to determine the features of IC curves, which have a strong relationship with battery capacity fading [24]. According to the definition of ICA, a common method to obtain the IC curve is the numerical-derivative method, and the charging capacity divided by terminal voltage change within the equal voltage interval (EVI) or equal time interval (ETI) is the IC value. However, the selected voltage or time interval has a severe impact on the IC curves; moreover, the IC curves are susceptible to the measurement noise [25]. Therefore, it is necessary to adopt a proper filter to obtain a smooth IC curve.

The Lorentzian function was adopted to fit the IC curves in ref [24], and the fitting results showed that the proposed model was capable of seizing the voltage plateaus as well as the IC peaks. Moreover, the model parameters had explicit meanings associated with electrode phase transition behaviors. Bian et al. [26] improved the capacity model in ref [24], and the current maximum capacity could be estimated directly by fitting the constant current (CC) charging curve. The problem is that more than ten parameters are unknown and need to be identified. The Gaussian filter also has superiority in smoothing charging data and was employed in Refs. [20,27]. In Ref. [20], the influence of sampling frequency on the IC curves was investigated, and the position of IC peaks was adopted to estimate battery capacity after obtaining the IC curves under different battery aging statuses. The experiment results showed that the maximum error of capacity estimation was 2.5%. Inspired by that support vector regression could smooth the IC curves [28], Feng et al. [29] made improvements for support vector regression in terms of algorithm speed, adaptability, and estimation accuracy, and the peak area was found to be a better indicator to estimate the battery capacity. Assuming that the measured voltage-charging capacity data during a small range can be described with a polynomial function, Wang et al. [30] proposed a central least squares (CLS) method to smooth the differential voltage (DV) curves and found it is a more robust method compared with the traditional numerical-derivative method. Zheng et al. [31] transferred the conventional voltage-based IC/DV curves to the SOC-based IC/DV curves and proposed a joint estimation method for battery SOC and capacity through combining three feature points on IC and DV curves. However, the feature points are too few to correct the battery SOC in time. After obtaining the battery IC curves at different aging statuses, the grey relational analysis [32] and the ridge regression [33] were utilized to establish the correlation between the battery SOH/capacity and the features of IC curves. To enhance the adaptability of ICA method in the battery module, Weng et al. [34] extended the ICA based capacity estimation from a single cell to the battery module, and the correlation between the normalized capacity and normalized peak using single-cell data was generalizable to parallel-connected cells with various aging statuses. Similar to the traditional ICA based on voltage and charging capacity, Samad et al. [35] innovatively adopted a mechanical rather than electrical signal and put forward a force-based ICA, which was with low signal to noise ratio and high estimation accuracy.

1.2. Existing challenges and original contributions

Although previous researches on capacity estimation based on the ICA method have proven its effectiveness, there still exist following critical challenges: (1) The computational complexity and efficiency are essential for vehicular online state estimation, especially for the generalization of ICA method to estimate battery capacity from single-cell to multi-dimension. The solution of the fitting functions for the charging curves needs to identify more than ten unknown parameters [22,24], which requires complex optimization algorithms. Besides, some existing smoothing methods for IC curves may suffer from the problem of time-consuming, as well as requiring a wide range of charging data to process the IC value at a time. The above issues challenge the online application of this method to estimate battery capacity; (2) The charging condition has a serve impact on the IC curves, while previous ICA researches mainly focus on the standard charging (charging from 0% SOC with a 0.5C current at 25 °C). In vehicle applications, it is rare to discharge the batteries fully; thus, the battery will be charged from non-zero initial SOC. Under these circumstances, the polarization process of battery charging from non-zero SOC is different from that of battery charging from 0% SOC and will lead to the deformation of the IC curves eventually. Apart from this, the battery internal resistance during charging will be influenced by battery temperature, which may cause the IC curve to shift. Therefore, the non-standard charging processes will challenge the adaptability of the traditional ICA method.

Aiming at the above problems, as the main contributions of this paper, a fast estimation of battery capacity based on ICA considering the charging condition is established. First, a smoothing method for IC curves based on KF is proposed in the paper, which requires less computation and far fewer data at a time to handle the IC value. What's more, in addition to the standard charging experiments to obtain the correlation between capacity and features of IC curves, some non-standard charging experiments are also performed, including the battery charging from different initial SOCs and charging at different temperatures, to investigate the influence of charging initial SOC and temperature on the IC curves. Finally, a correction method considering initial SOC is established for capacity estimation, which is with high accuracy and robust to the initial SOC error.

1.3. Outlines of the paper

The remainder of this paper is organized as follows: Section 2 introduces the acquisition of the IC curves based on the proposed KF method. Section 3 explains the battery experiments with standard and non-standard charging. In Section 4, the effectiveness of the proposed smoothing method is verified, then the influence of battery aging statuses, charging initial SOC, and temperature on the IC curves is investigated. Finally, the adaptive capacity estimation based on ICA is proposed for lithium-ion batteries. Some conclusions and summarizations are provided in the final section.

2. IC curves acquisition

2.1. The derivation of IC curves

Fig. 1(a) shows the curves of charging capacity to battery voltage at different SOHs during the CC pattern (charging current is 0.5C). In this paper, the definition of SOH is the ratio of current maximum capacity to the nominal capacity [36]. The charging curves have a downward trend with the attenuation of battery capacity. Besides, it can be found that the charging curves have different slopes within different voltage ranges. For example, for the battery with 99.67% SOH, the slope of the charging curve between 3.52 V and 3.68 V has changed because of the phase transition process. An inapparent phenomenon is that batteries with different SOHs have different slope changes.

The principle of the ICA method is to calculate the derivative of the charging capacity to battery voltage, or in other words, obtaining the changing rate of charging capacity at a specific voltage. The voltage plateaus of the charging curve of the battery with 99.67% SOH (Fig. 1(a)) can be converted to peaks on the IC curve (Fig. 1(b)). Fig. 1(b) demonstrates three peaks on the IC curve, which are located at around 3.52 V, 3.68 V, and 4.0 V, respectively. Generally, the position, height, and area of the IC peaks are considered as the IC features, and the relationship between the IC features and capacity attenuation needs to be further investigated.

The acquisition of IC curves through numerical-derivative method (EVI and ETI) can adopt the following equation:

$$\begin{cases} \mathrm{EVI{:}IC} = \frac{dQ_c}{dV_c} \approx \frac{\Delta Q_c}{\Delta V_c} = \frac{Q_{c,2}-Q_{c,1}}{\Delta V_c} \\ \mathrm{ETI{:}IC} = \frac{dQ_c}{dV_c} \approx \frac{\Delta Q_c}{\Delta V_c} = \frac{i_c \Delta t}{V_{c,2}-V_{c,1}} \end{cases} \tag{1}$$

where Q_c is the battery charging capacity, V_c and i_c are the battery terminal voltage and current during the CC charging pattern, respectively. For the EVI method, the IC is calculated every equal voltage change, and $Q_{c,2} - Q_{c,1}$ represents the variation of charging capacity during the interval; For the ETI method, the IC is calculated every equal time change Δt, and $V_{c,2} - V_{c,1}$ represents the voltage change during the interval. No matter which method is adopted, the IC curve will be severely affected by the selected interval. If the interval is small, the IC curve will be noisy, and if the interval is large, the IC curve features will become indistinct. Thus, a more adaptive data smoothing method shall be adopted.

2.2. The KF method to smooth the IC curves

In this paper, we propose an adaptive processing method based on the KF for the IC curves. The IC value at the kth sample point is defined as $(\frac{dQ_c}{dV_c})_k$, and the measured battery voltage and current satisfy the

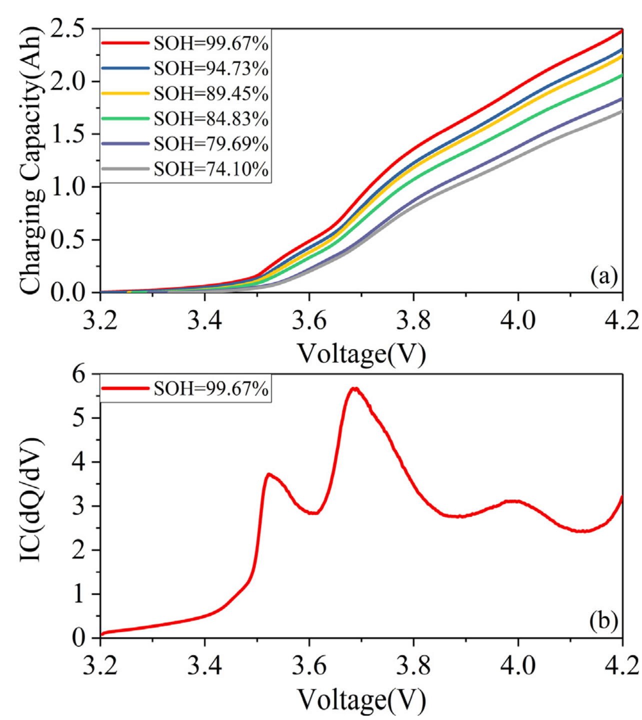

Fig. 1. The diagram of ICA: (a) Battery voltage-charging capacity curves at different SOHs; (b) The IC curve of the battery with 99.67% SOH.

following equation:

$$\begin{cases} (\frac{dQ_c}{dV_c})_{k+1} = (\frac{dQ_c}{dV_c})_k + w_k \\ \int_{k-l_{KF}/2}^{k+l_{KF}/2} i_c \mathrm{d}t = (V_{c,k+l_{KF}/2} - V_{c,k-l_{KF}/2}) \times (\frac{dQ_c}{dV_c})_k + v_k \end{cases} \quad (2)$$

where $(\frac{dQ_c}{dV_c})_{k+1}$ represents the IC value at the $(k+1)$th sample point, l_{KF} is data length to process the IC value at a time, $\int_{k-l_{KF}/2}^{k+l_{KF}/2} i_c \mathrm{d}t$ represents the charging capacity from $(k - l_{KF}/2)$th sample point to $(k + l_{KF}/2)$th sample point. w_k and v_k are the process and measurement noise, and are both considered as Gaussian white noise with zero mean and covariance Q and R, separately. The Eq. (2) is derived from the ETI method shown in Eq. (1).

The optimal solution of $(\frac{dQ_c}{dV_c})_k$ can be estimated by some adaptive filters, for example, KF [37,38], H_∞ filter [39], and particle filter [40], which have also been widely employed in battery state estimation. In this paper, we adopt the KF and the detailed procedures of the KF algorithm are shown in Table 1.

where x_k, u_k, and y_k are the system state, input, and output, respectively. In this paper, there is no system input, and system state and output are $x_k = (\frac{dQ_c}{dV_c})_k$ and $y_k = \int_{k-l_{KF}/2}^{k+l_{KF}/2} i_c \mathrm{d}t$. Accordingly, the input matrix B_k and feedthrough matrix D_k are zero matrices, and the system and output matrixes are $A_k = 1$ and $C_k = (V_{c,k+l_{KF}/2} - V_{c,k-l_{KF}/2})$, separately. $\hat{x}_k$ is the estimated value of x_k. $\hat{x}_k^-$ and P_k^- are the *a priori* predicted value of the system state and its error covariance, while $\hat{x}_k^+$ and P_k^+ are the *a posteriori* updated value according to the measurement information.

Based on the above KF algorithm, the battery IC curves will be obtained smoothly and expeditiously.

3. Experiment setups and design

To obtain the relationship between IC features and capacity attenuation, battery charging experiments shall be designed and conducted. A kind of commercial NCM/graphite cylindrical battery with a rated capacity of 2750 mAh and a nominal capacity of 2900 mAh is used in this study. The battery charge and discharge cut-off voltage are 4.2 V and 2.5 V, respectively. The recommended charging temperature ranges from 0 °C to 45 °C. The experiment setups include a battery charge & discharge system, a temperature measurement station, an environmental chamber, and a host computer.

1 Battery standard charging experiments

The standard charging means that the battery is charging from 0% SOC at 25 °C. The charging data is selected from the capacity tests during the routine operation experiments and accelerated aging experiments. Usually, the capacity test includes charging to 100% SOC through the constant current-constant voltage (CC-CV) pattern and discharging to 0% SOC, and 1 h rest between charging and discharging. The capacity test is repeated at least three times. If the deviation between the maximum and minimum discharging capacity is less than 1.5%, the average discharging capacity in several times is considered as the current maximum available capacity, and the data during the charging process is selected as battery standard charging. The sampling frequency during standard charging is 10 Hz.

2 Battery non-standard charging experiments

The battery non-standard charging experiments include the battery charging from different initial SOCs and charging at different temperatures, in order to investigate the influence of these factors on the IC curves.

In the experiments of charging from different initial SOCs, eight cells (No.1–No.8) with different aging statuses involve in the experiments. Every 10% SOC is taken into account, and the detailed experimental procedures are shown in Table 2. The experiments are conducted at 25 °C, and the sampling frequency during charging is 10 Hz.

In the experiments of charging at different temperatures, six cells (No.1–No.6) with different aging statuses involve in the experiments. It is noted that only the experiment of CC pattern is taken into account and battery charges within the recommended temperature range. The detailed experimental procedures are shown in Table 3, and the sampling frequency during charging is 10 Hz.

4. Results and discussion

4.1. The effectiveness of KF method to acquire the IC curves

Wang et al. [30] proposed a CLS method to handle the battery charging data and obtain the DV curve, which adopting a one-order polynomial function to fit the data sequence. In this section, the effectiveness of the traditional CLS and the proposed KF method to process the IC curves will be compared. An essential parameter in the CLS method is the data length l_{CLS}, and a short data length will make the IC curves noisy, while the features of the IC curves will become indistinct if the data length is considerable. Fig. 2(a) shows the IC curves obtained through the CLS method with different data lengths. It can be seen that the larger the data length, the smoother the IC curve. When $l_{CLS} = 100$ (about 10 s), the obtained IC curve is extremely noisy, and the features of the IC curves related to the battery aging cannot be accurately identified, and also when the data length is 200. When l_{CLS}= 500 and 1000, the obtained IC curves become much smoother. From the enlarged part of Fig. 2(a), it can be confirmed that the smoothness of the IC curve with $l_{CLS} = 1000$ is slightly better than that with $l_{CLS} = 500$. Although the IC curve with the data length of 1000 is still a little volatile, the features of the IC curve are distinct. Therefore, if the CLS method is adopted to smooth the IC curves, the data length of 1000 is suitable.

In this paper, we adopt the KF method to smooth the IC curve. Before employing the KF method, some important initial parameters shall be determined, including the initial state and error covariance, and the covariance of the process and measurement noise. Here, we set $\hat{x}_0^+ = 0$, $P_0^+ = 1$, and $Q = R = 10^{-6}$. Fig. 2(b) shows the IC curves obtained from the KF method with different data lengths, and obviously, the KF method can still generate a smooth IC curve even if the data length is 50. Moreover, it can be found that the KF method with a small data length (for example, l_{KF}= 50 and 100) reduces the peak value, and moves the IC curves slightly to the high voltage. The IC curve with $l_{KF} = 10$ (not shown in Fig. 2(b)) is also obtained; however, the features of the IC curve are distorted, especially the first peak. It is because that

Table 1
Algorithm of the KF.

Linear system		
State and output equation:	$\begin{cases} x_{k+1} = A_k x_k + B_k u_k + w_k \\ y_k = C_k x_k + D_k u_k + v_k \end{cases}$	(3)
Initialization		
State and error covariance:	$\begin{cases} \hat{x}_0^+ = \mathrm{E}(x_0) \\ P_0^+ = \mathrm{E}[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T] \end{cases}$	(4)
Process and measurement noise:	$\begin{cases} Q = \mathrm{E}(w_k w_k^T) \\ R = \mathrm{E}(v_k v_k^T) \end{cases}$	(5)
Recursive calculation		
State and error covariance time update:	$\begin{cases} \hat{x}_k^- = A_k \hat{x}_{k-1}^+ + B_k u_k \\ P_k^- = A_k P_{k-1}^+ A_k^T + Q \end{cases}$	(6)
Kalman gain:	$K_k = P_k^- C_k^T (C_k P_k^- C_k^T + R)^{-1}$	(7)
State and error covariance measurement update:	$\begin{cases} \hat{x}_k^+ = \hat{x}_k^- + K_k(y_k - C_k \hat{x}_k^- - D_k u_k) \\ P_k^+ = (I - K_k C_k) P_k^- \end{cases}$	(8)

Table 2
Battery non-standard charging from different initial SOCs.

Step	Experiment Description
1	Set the chamber temperature to 25 °C
2	Rest the battery for 1 h, and discharge the battery to 10% SOC with 0.5C current
3	Rest the battery for 1 h, and charge the battery to 100% SOC in CC-CV mode with 0.5C current
4	Rest the battery for 1 h
5	Discharge the battery to 20%, 30%, 40%, 50%, 60%, 70%, 80% and 90%, repeat Step 3 to Step 4
6	End

the output matrix C_k is severely affected by the voltage measurement noise within such a small data length (about 1 s). This problem can be solved by adjusting the covariance of the process and measurement noise (increasing Q or decreasing R). The IC curves with l_{KF}= 200 and 500 are much smoother and almost coincide with each other; therefore, these data lengths can be adopted in the acquisition of IC curves.

Fig. 2(c) shows the comparison of the CLS and KF methods to obtain the IC curves. It can be confirmed that the IC curve obtained from the CLS method with l_{CLS} = 1000, and that obtained from the KF method with l_{KF}= 200 and 500 almost overlap, in other words, the KF and the CLS method have the same effect on processing the IC curves, while the KF method requires far fewer data at a time to handle the IC value.

Table 4 shows the comparison of the computational efficiency of CLS and KF methods to acquire the full IC curve from the charging curve with 64,690 points. Both the CLS and KF methods are carried on a personal computer (Intel Core i7 8550, CPU 1.8 GHz) with the Matlab environment. It is noted that the final computational time of each method is calculated from the average of ten runs' computational time. With the increasement of the data length, both the computational time of the CLS and KF methods raise. Using the same data length to handle the IC curves, the KF method saves at least 80% more than the CLS method. To acquire a smooth IC curve, the recommended value of l_{CLS} is 1000, and the computational time is 2.101 s, while for the KF method, the recommended value of l_{KF} is 200, and the computational time is 0.191 s. The time saving is tremendous; therefore, in the following analysis, the data length for the KF method is selected as 200.

The above analysis proves the effectiveness and rapidity of the KF method to process the IC curve. The KF method can generate a smooth IC curve with a saving of more than 80% of the computational time. Hence, this method is promising to handle the capacity estimation on-board, especially for hundreds of cells in series.

4.2. Battery IC curve characteristics

4.2.1. The IC curves during standard charging at different aging statuses

According to test specifications for battery standard charging, a total of 191 standard charging tests during the routine operation and accelerated aging experiments are selected for the investigation of ICA.

Fig. 3 shows the IC curves of the battery at different SOHs. It is evident that, for different battery aging statuses, the IC curves generated through the KF method are smooth and without much noise, and the features of the IC curves are distinct, which can be used in ICA. Similar to the IC curves of the NCM battery shown in Ref. [20], the IC curves in this paper demonstrate three peaks (Peak A–C), which initial position is around 3.52 V, 3.68 V, and 4.0 V. The difference is that within 4.0–4.2 V, the IC value decreases first and then increases, resulting in the appearance of Valley C in this paper. However, in Refs. [29,41], the IC curves of battery with NCM cathode only show two distinct peaks, and the third peak is hard to be identified. This may be due to the different ratios of electrode materials produced by different manufacturers, and the charging current also has a significant influence on the IC curves.

From 99.67% SOH to 79.69% SOH, it can be confirmed that three peaks can be distinctly detected. However, for the battery with 74.10% SOH, the Peak A and Valley A mix together and cannot be distinguished. With the attenuation of the battery capacity, the height of Peak A, B, and C decreases, and the reduction of Peak A and C is much smaller than that of Peak B. The reduction of IC peaks is closely related to the happening of loss of active material during battery aging [21]. In addition, the position of these peaks remains unchanged at a high SOH, while the IC curve has a moving trend to the high voltage at a low SOH. Through the comparison of degradation modes shown in Ref. [19], the most pertinent aging mechanism for this phenomenon may be that the loss of lithium inventory happens. For an exhaustive deduction of battery degradation mechanism, the charging/discharging process with extreme low current is essential for the ICA. Generally, the transformation of these IC features, especially the height of peak and valley, has a close relationship with battery capacity attenuation and can be employed to perform the capacity estimation.

4.2.2. The IC curves with different charging initial SOCs

During actual vehicle application, it is rare to discharge the battery to 0% SOC, and the initial SOC will have a significant influence on battery charging curves. Fig. 4 shows the curves of battery voltage-remaining capacity during CC pattern with different initial SOCs. Here the remaining capacity means the sum of residual capacity before charging and the charging capacity. At the beginning of charging, the charging curve with non-zero SOC does not overlap with the standard charging curve (initial SOC = 0%) due to the polarization effect, and then the difference between the non-standard and standard charging curves gradually decreases. When the battery voltage is larger than 3.9 V, the charging curves with the initial SOC less than or equal to 60% almost coincide with each other.

Fig. 5 shows the IC curves of the battery with different initial SOCs at different aging statuses (SOH = 99.18%, 88.12%, and 79.51%). Obviously, the initial SOC seriously affects the IC curves, and the IC curves will be distorted if the battery is not charging from 0%. For the battery with 99.18% SOH shown in Fig. 5(a), because of increasement of the battery voltage at the beginning of charging, only the IC curves with initial SOC = 0% and 10% have the feature of Peak A, and the height of Peak A with initial SOC = 10% shows a sharp decline. For the IC curves with initial SOC = 0–50%, the Peak B can be identified; however, this peak is also severely reduced and has a moving trend to

Table 3
Battery non-standard charging at different temperatures.

Step	Experiment Description
1	Set the chamber temperature to 25 °C
2	Rest the battery for 1.5 h, and discharge the battery to 2.5 V with 0.5C current
3	Set the chamber temperature to 0 °C
4	Rest the battery for 1.5 h, and charge the battery to 4.2 V with 0.5C current
5	Set the chamber temperature to 25 °C
6	Rest the battery for 1.5 h, and discharge the battery to 2.5 V with 0.5C current
7	Set the chamber temperature to 10 °C, 35 °C and 45 °C, repeat Step 4 to Step 6
8	End

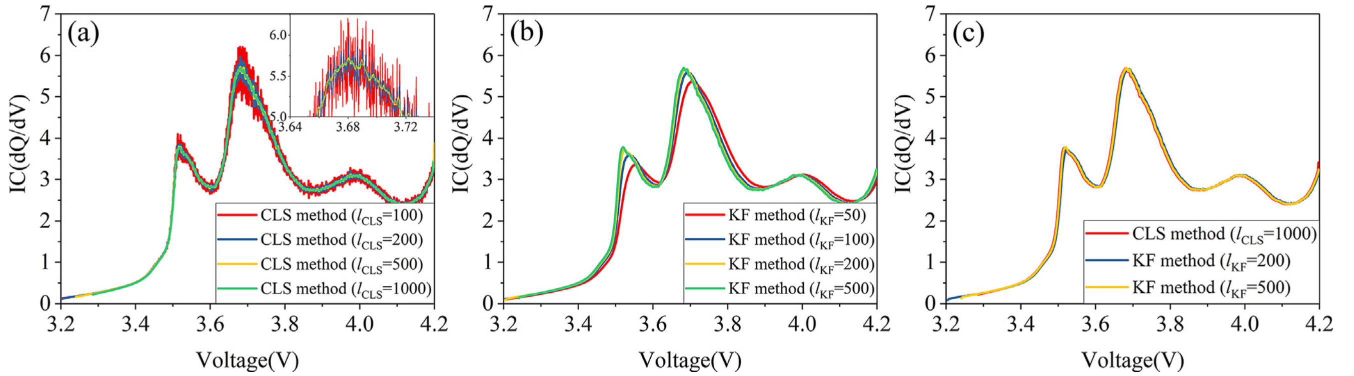

Fig. 2. Battery IC curves: (a) The IC curves obtained from CLS method with different data lengths; (b) The IC curves obtained from KF method with different data lengths; (c) The comparison of the CLS and KF methods to smooth the IC curves.

Table 4
Comparison of the computational efficiency of CLS and KF methods.

The CLS method	l_{CLS}	100	200	500	1000
	Computational Time(s)	1.026	1.184	1.504	2.101
The KF method	l_{KF}	50	100	200	500
	Computational Time(s)	0.153	0.161	0.191	0.262

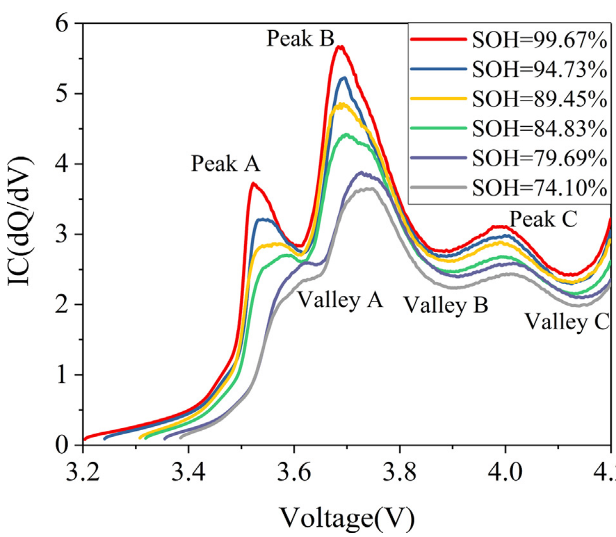

Fig. 3. The IC curves at different battery aging statuses.

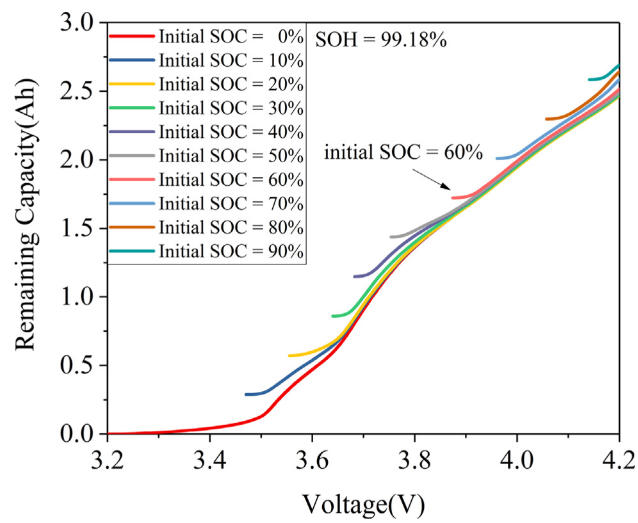

Fig. 4. Battery voltage-remaining capacity curves with different charging initial SOCs.

high voltage. The Peak C will disappear if the initial SOC is larger than 70%, while the height and position of Peak C seem to be unchanged when the initial SOC ranges from 0% to 60%, which is promising to be employed in capacity estimation in the partial charging.

The IC curves of the aged batteries shown in Fig. 5(b) and (c) present similar characteristics. The difference is that the height of Peak C is reduced with the increasement of initial SOC for the battery with 88.12% and 79.51% SOH. From the enlarged parts of Fig. 5(b) and (c), it can be found that the lower the battery SOH, the higher the influence of initial SOC on the height of Peak C. Therefore, it concludes that the height of Peak C is affected by the initial SOC and battery aging status together, which can be considered as an essential indicator during non-standard charging.

4.2.3. The IC curves at different temperatures

Although battery thermal management could adjust the battery temperature within a suitable range, there is still a deviation with the ambient temperature (25 °C). Fig. 6 shows the battery charging curves during the CC pattern at different temperatures. It is evident that with increasement of battery temperature, the overall charging capacity during CC pattern increases. When the temperature decreases, the charging curve moves to the right caused by the increasement of battery internal resistance. The relationship between battery temperature and resistance can be described by the Arrhenius equation reliably [42], which can explain the phenomenon that the distance between charging curves at 0 °C and 25 °C is much greater than that between the charging curves at 25 °C and 45 °C.

Fig. 7 shows the IC curves of the battery at different temperatures, and the corresponding battery SOH is 98.81%, 92.07%, and 81.20%, respectively. Similar to Fig. 6, the temperature will influence the position of the IC curves. With the decrease of battery temperature, the IC curve moves to the high voltage. Furthermore, it can be found that the Peak C of the battery with 81.20% SOH at 0 °C cannot be identified clearly due to curve moving. Refs. [43,44] also showed the IC curves at different temperatures, and both satisfied the characteristic that low temperature would cause IC curve to shift to high voltage. The height of the IC curves is also influenced by battery temperature; however, it seems to be irregular with temperature. Generally, with the decrease of temperature, the Peak A becomes higher, and Peak C decreases, while the height of Peak B remains unchanged approximately. Through the comparison of Figs. 3 and 7, the influence of temperature on the height of Peak B is much smaller than that of battery aging. Therefore, it can be approximately assumed that the height of peak B is not affected by temperature and can be employed to estimate the battery capacity at different temperatures.

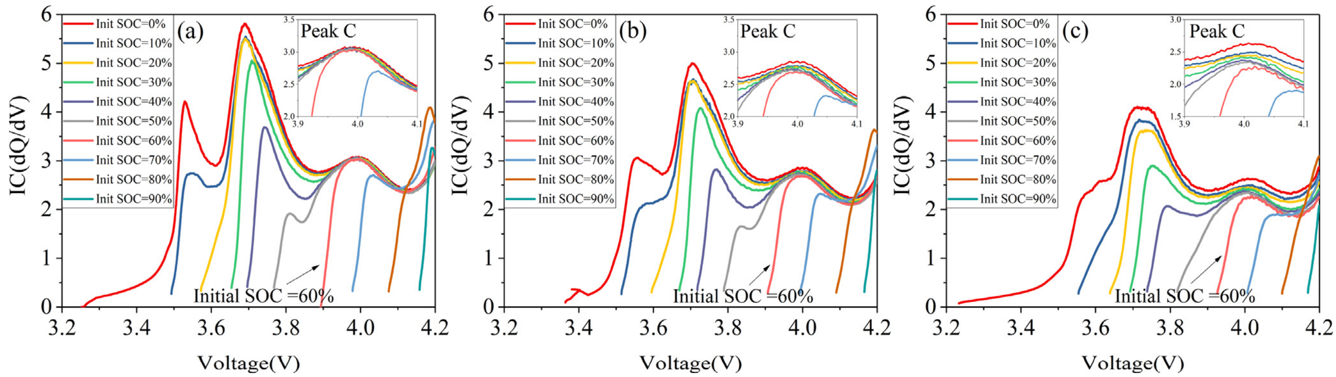

Fig. 5. IC curves of battery with different charging initial SOCs at different aging statuses: (a) Battery with 99.18% SOH; (b) Battery with 88.12% SOH; (c) Battery with 79.51% SOH.

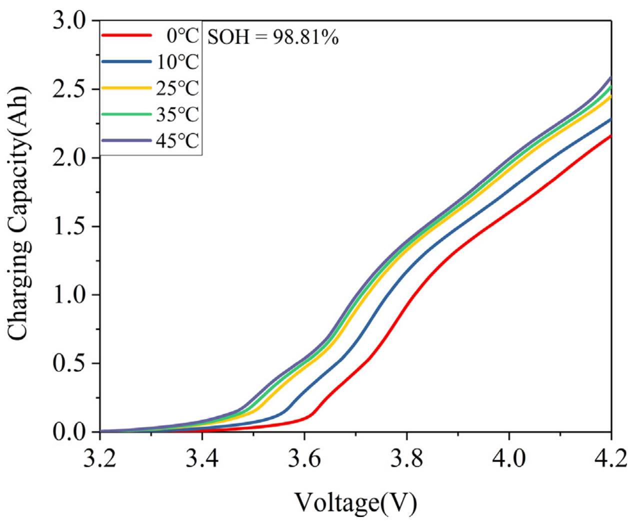

Fig. 6. Battery charging curves at different temperatures.

4.3. Capacity estimation based on ICA

4.3.1. Capacity estimation during standard charging

The position, height, and area of the peaks are commonly employed to evaluate capacity attenuation. In this paper, we will adopt the height of the IC peaks and valleys to estimate the battery capacity.

Because Peak A and Valley A cannot be distinguished when the battery is in a severe aging status, these two features are out of consideration here. Fig. 8 shows the correlation between the height of Peak/Valley B and C and battery capacity. It can be confirmed that, except for the Valley C shown in Fig. 8(d), the height of Peak B, Valley B, and Peak C has a strong correlation with the battery capacity. The linear fitting lines and the coefficient of determination (R^2) are also presented in Fig. 8, and for Peak B, Valley B, and Peak C, R^2 are all greater than 0.95. Besides, almost all the feature points are within the ± 3% error boundaries except for a few outliers in Fig. 8(a), (b), and (c). Hence, the height of the Peak B, Valley B, and Peak C will be employed in the ICA method. Moreover, the correlation is adaptive to different batteries (No.1–No.9), which enhances the promise of the ICA method to estimate the battery capacity. It can also be confirmed that the position of Peak B, Valley B, and Peak C under different battery aging statuses are within 3.670–3.745 V, 3.875–3.925 V, and 3.970–4.025 V, respectively. Therefore, the determination of the height of the peak/valley can be performed in these voltage ranges to identify the features correctly, as well as reduce the computation load.

The linear relationship between the battery capacity and the features can be established with the following equation:

$$C_{F_i} = \alpha_i \times H_{F_i} + \beta_i \tag{9}$$

where H_{F_i} represents the identified height of the ith feature (1st is Peak B, 2nd is Valley B, and 3rd is Peak C), C_{F_i} represents the estimated capacity according to the ith feature. The corresponding α_i and β_i for capacity estimation are listed in Table 5. For the standard charging, three estimated values for battery capacity will be obtained, and the average or weighted average of these values can be considered as the final estimated capacity. The weighted coefficient can depend on the R^2 of each linear regression.

Table 6 lists the estimated capacity through the ICA method during standard charging. Four cells are selected to verify the ICA method, including No.9–No.12. The actual capacity of these cells is also listed in Table 6. According to identified H_{F_1}~H_{F_3}, three capacity values are obtained based on the Eq. (9), and it can be found that there are deviations between C_{F_1} and C_{F_3}. The final estimated capacity for each cell is the weighted average of C_{F_1}~C_{F_3}, and it can be confirmed that the final estimated capacity is close to the actual capacity. The maximum

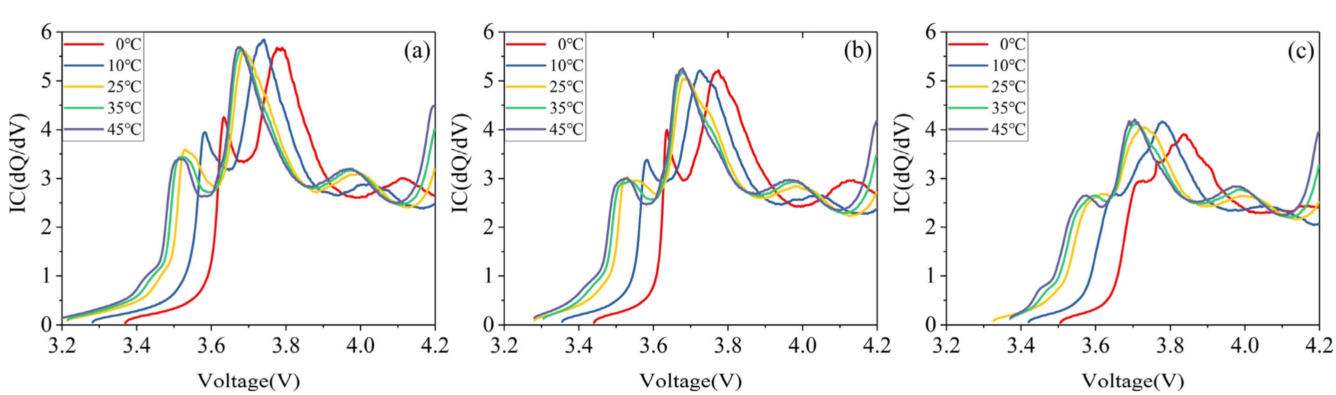

Fig. 7. IC curves of battery at different temperatures: (a) Battery with 98.81% SOH; (b) Battery with 92.07% SOH; (c) Battery with 81.20% SOH.

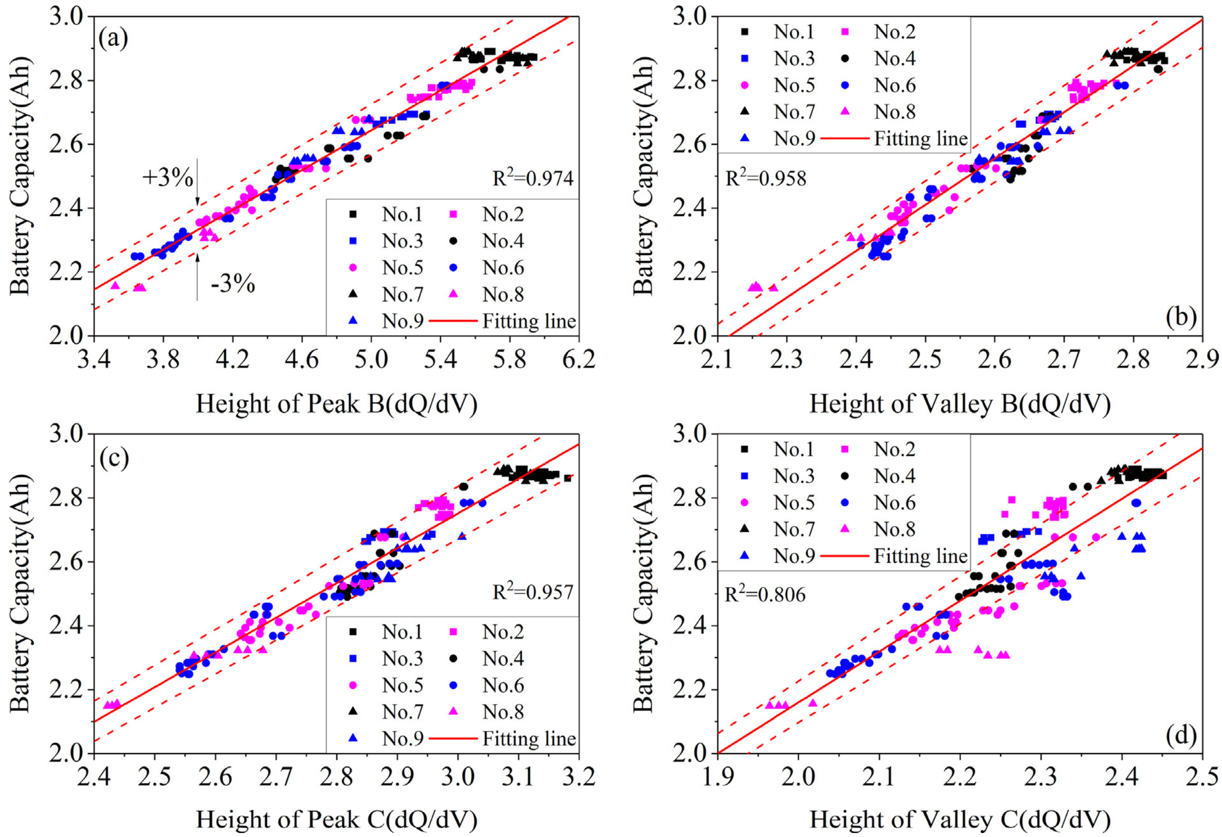

Fig. 8. The correlation between the features of IC curves and battery capacity: (a) Peak B; (b) Valley B; (c) Peak C; (d) Valley C.

Table 5
Coefficients for the capacity estimation with different features.

Feature	i	α_i	β_i
Peak B	1	0.3118	1.0857
Valley B	2	1.4502	−1.1416
Peak C	3	1.0878	−0.5113

absolute estimation error of these four cells is 2.092%, which proves the effectiveness of the ICA method.

4.3.2. Capacity estimation considering the charging initial SOC

The height of Peak C is an important indicator during battery non-standard charging at 25 °C. The correlation between the battery capacity and the height of Peak C with different initial SOCs ranging from 10% to 60% is demonstrated in Fig. 9. The situation that SOC larger than 60% is out of consideration in this paper because of deformation or disappearance of Peak C. It can be confirmed that no matter the what the initial SOC is, the battery capacity always has a strong linear relationship with the height of Peak C, and all the R^2 are greater than 0.96. Moreover, the correlation is adaptive to different batteries (No.1–No.8); hence, this correlation is universal.

The coefficients for the linear regression shown in Fig. 9 are listed in Fig. 10. It should be noted that, although the initial SOC is expected to be adjusted to a certain value (for example, 10%), it still has some deviations (from 9.72% to 11.62% in several experiments); therefore, the average value of these SOCs is taken into consideration here. As shown in Fig. 10, with the increasement of initial SOC, the coefficient $\alpha_{3,\,SOC}$ decreases while the coefficient $\beta_{3,\,SOC}$ increases. Combining the coefficients for Peak C (standard charging) listed in Table 5, the relationship between coefficients for Peak C and initial SOC ranging from 0% to 60% can be fitted by a linear function shown in the Eq. (10). Moreover, the fitness of the linear function is acceptable, and R^2 for $\alpha_{3,\,SOC}$ and SOC is 0.9381, while for $\beta_{3,\,SOC}$ and SOC is 0.9505. Therefore, it is promising to correct the regression coefficients between battery capacity and the height of Peak C according to the initial SOC.

$$\begin{cases} \alpha_{3,SOC} = -0.00572 \times SOC + 1.05348 \\ \beta_{3,SOC} = 0.01791 \times SOC - 0.40996 \end{cases} \tag{10}$$

In conclusion, the correction model considering the charging initial SOC to estimate battery capacity is constructed, and the detailed adaptive capacity estimation method based on ICA is illustrated in Fig. 11. The main advantage of this adaptive method is that the ICA based capacity estimation takes the charging initial SOC into consideration and selects the height of Peak C to correct the non-standard charging ICA.

Two cells (No.9 and No.10) with different aging statuses are

Table 6
Capacity estimation results of the ICA method during standard charging.

Battery No.	Actual Capacity (Ah)	H_{F_1}	H_{F_2}	H_{F_3}	C_{F_1}(Ah)	C_{F_2}(Ah)	C_{F_3}(Ah)	Estimated Capacity (Ah)	Estimation Error (%)
No.9	2.555	4.7291	2.5819	2.8903	2.5602	2.6028	2.6328	2.5984	1.689
No.10	2.834	5.6480	2.7307	3.1083	2.8467	2.8186	2.8699	2.8451	0.400
No.11	2.631	4.8818	2.5636	2.8073	2.6078	2.5762	2.5425	2.5757	−2.092
No.12	2.808	5.5913	2.7008	2.9946	2.8290	2.7752	2.7463	2.7838	−0.848

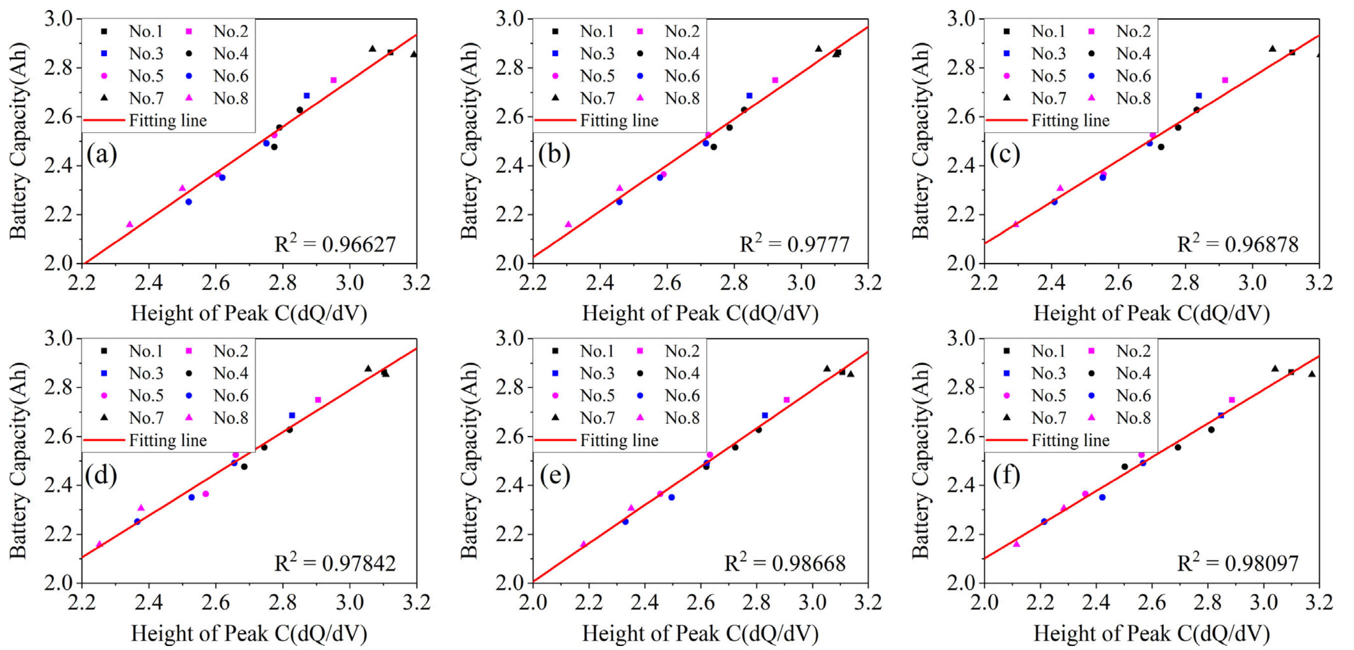

Fig. 9. The correlation between the battery capacity and height of Peak C with different initial SOCs: (a) Initial SOC = 10%; (b) Initial SOC = 20%; (c) Initial SOC = 30%; (d) Initial SOC = 40%; (e) Initial SOC = 50%; (f) Initial SOC = 60%

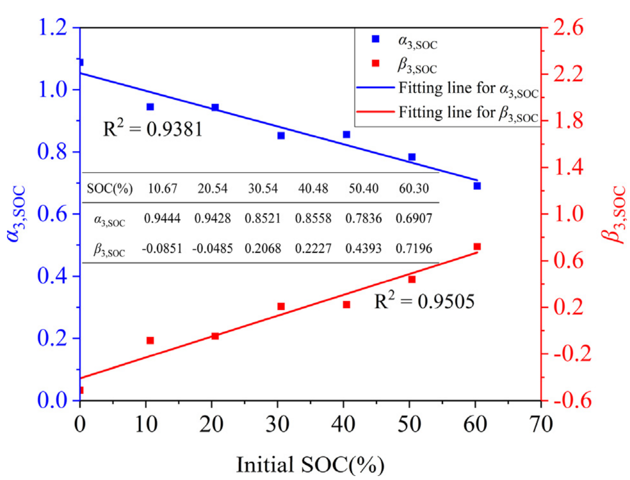

Fig. 10. Coefficients for the linear regression with different initial SOCs.

employed to verify the proposed method, and the actual capacity of these two cells is listed in Table 6. The identified Peak C based on the KF method is listed in Table 7. It can be concluded that no matter the battery aging statuses is, the position of Peak C remains approximately unchanged. However, the height of Peak C is affected by the battery aging status and charging initial SOC. For the battery with more severe aging (No.9), the height of Peak C significantly decreases with the increasement of charging initial SOC, while for the battery with less capacity attenuation (No.10), the height of Peak C has little change. This is similar to the characteristic shown in Fig. 5.

Fig. 12 shows the comparison of the proposed and traditional method; here, the proposed method refers to the method based on ICA considering the initial SOC and traditional method employs Table 5 to estimate capacity without considering initial SOC. It can be confirmed that, for Cell No.9 with severe aging, the proposed method is effective. Although the estimated capacity reduces slightly with the increasement of charging initial SOC, the relative capacity error is still within an acceptable range. The maximum deviation of the proposed method is −2.165% when the charging initial SOC is 60.19%. However, under the influence of initial SOC, the traditional method is powerless to estimate an accurate capacity. Although the estimation error is smaller than 3% when charging initial SOC is less than or equal to 30.32%, it is up to −10.046% when the initial SOC is 60.19%. Except for the initial SOC of 10.54%, the accuracy of the proposed method is higher than that of the traditional method, and the estimation error of the proposed method is reduced by 7.881% when the initial SOC is 60.19%.

For Cell No.10 with less capacity attenuation, the maximum deviations of the proposed and traditional method are 1.578% and 1.641%, respectively. It means that the traditional method can also obtain relatively accurate estimated capacity without considering the charging initial SOC when the battery with a high SOH. In general, the accuracy of the proposed method is slightly higher than that of the traditional method except for the initial SOC of 40.65%.

The above analysis shows the effectiveness of the proposed method, however, another problem needs to be paid attention to is the inaccuracy of the SOC estimation because the proposed method employs the initial SOC to correct the ICA model. Usually, the initial SOC can be determined by the OCV, and the SOC estimation error through this method is smaller than 1.2% after enough rest time [45]. Table 8 shows the estimated capacity of the proposed method with an inaccurate charging initial SOC. The SOC error of ± 5% is selected to verify the robustness of the proposed method. From Table 8, it can be found that, compared with the estimation without SOC error, the estimated capacity decreases when the SOC error is −5%, and that increases when the SOC error is 5%. In general, the absolute value of estimated capacity error is within 3% when the proposed method is disturbed by the SOC error of ± 5% , which extends the adaptability of the proposed method.

In all, the features of the IC curves have a close relationship with battery capacity. The proposed method considering the charging initial SOC shows high accuracy and is robust to the initial SOC error, which is promising to be an effective battery capacity estimation method for vehicle application.

5. Conclusions

This paper introduces a fast acquisition method of IC curves, then characteristics of IC curves during standard and non-standard charging are investigated; finally, the adaptive estimation based on ICA is proposed. Some conclusions and summarizations can be drawn below:

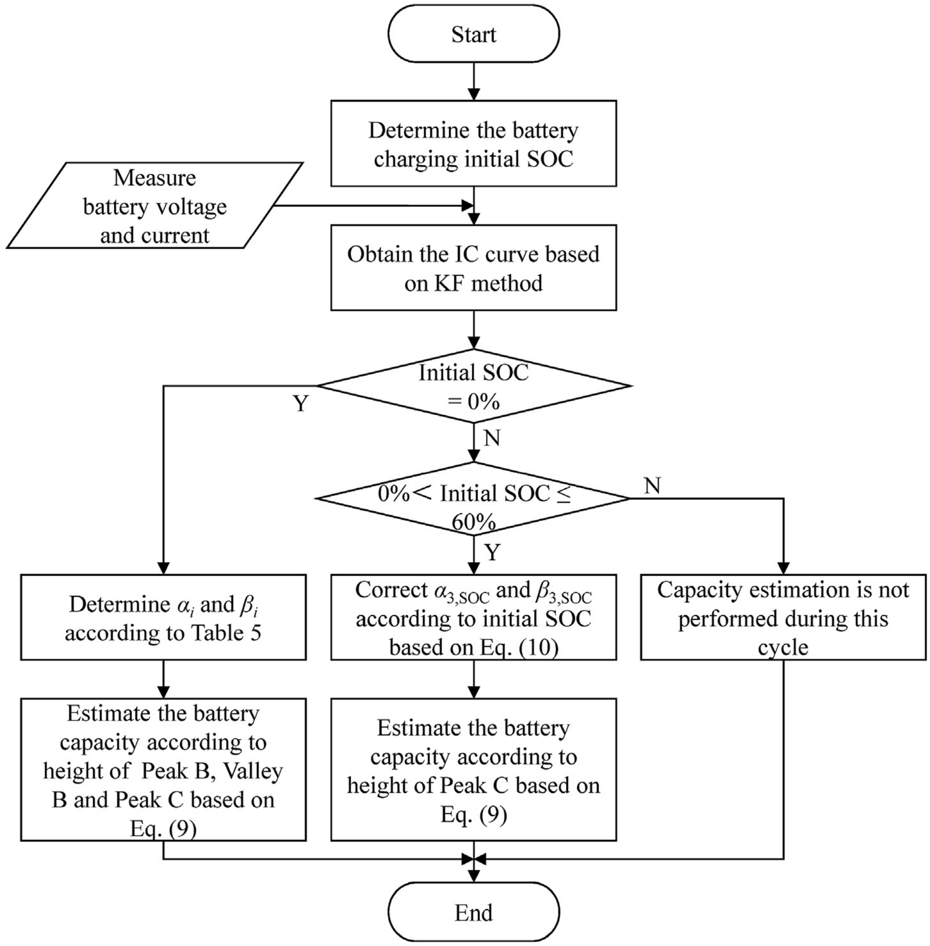

Fig. 11. Adaptive capacity estimation method based on ICA considering the charging initial SOC.

Table 7
The identified Peak C with different charging initial SOCs.

Battery No.	Charging initial SOC (%)	Position of Peak C (V)	Height of Peak C
No.9	10.54	4.010	2.8091
	20.42	3.994	2.7745
	30.32	4.002	2.7505
	40.21	4.010	2.7088
	50.23	3.993	2.6682
	60.19	4.010	2.5830
No.10	10.95	3.996	3.0424
	20.84	4.008	3.0356
	30.81	4.009	3.0647
	40.65	3.988	3.0920
	50.77	3.988	3.1178
	60.62	3.996	3.0630

(1) The KF algorithm is employed to process the IC curves. Compared with the traditional ICA smoothing method, the proposed smoothing method requires far fewer data at a time to handle the IC value. Moreover, the proposed method saves more than 80% of the computational time in obtaining a smooth IC curve.
(2) The experiments for battery standard and non-standard charging are designed and conducted. During the standard charging, the IC curve demonstrates three peaks and has a downward trend when battery capacity decreases. The battery charging initial SOC has a significant impact on the battery IC curve. When the battery is charging from non-zero SOC, the IC curve will be deformed, and some features will disappear. With the decrease of charging temperature, the IC curve moves to the high voltage.
(3) The height of Peak B, Valley B, and Peak C has a strong relationship with the capacity attenuation, and all the R^2 are greater than 0.95. Three features during standard charging are employed to estimate battery capacity, and the estimation results show the effectiveness of the ICA method. The height of Peak C always has a strong relationship with battery capacity when the charging initial SOC is less than or equal to 60%. The adaptive method adopts the charging initial SOC to correct the capacity coefficients, combined with the height of Peak C, to estimate battery capacity. The experimental results show the effectiveness of the proposed method, especially for the battery with severe aging. Moreover, the proposed method is robust to the initial SOC error, and the absolute value of capacity estimation error is within 3% when the proposed method is disturbed by the $\pm 5\%$ initial SOC error.

CRediT authorship contribution statement

Bo Jiang: Conceptualization, Methodology, Software, Data curation, Formal analysis, Validation, Writing - original draft, Writing - review & editing. **Haifeng Dai:** Methodology, Investigation, Funding acquisition, Validation, Writing - original draft, Writing - review & editing. **Xuezhe Wei:** Project administration, Resources, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to

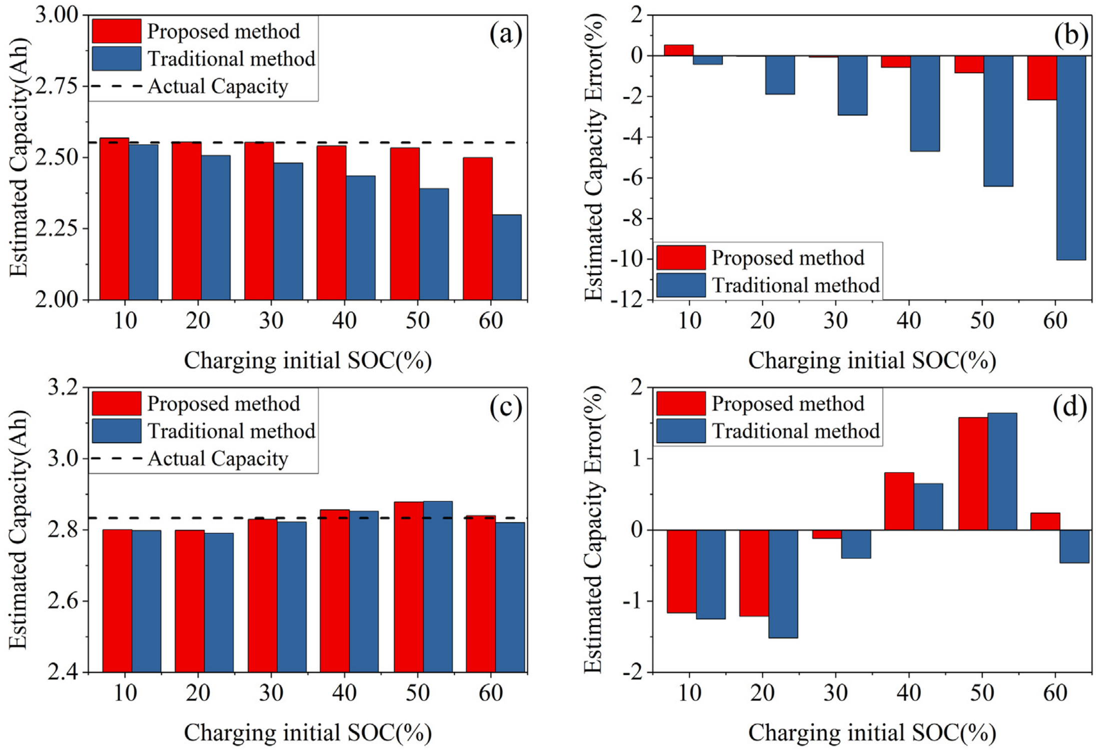

Fig. 12. Comparison of the proposed and traditional method: (a) and (c) are estimated capacity for Cell No.9 and No.10; (b) and (d) are the relative estimated capacity error for Cell No.9 and No.10.

Table 8
Capacity estimation for cell No.9 with inaccurate charging initial SOC.

Initial SOC Error1 (%)	Inaccurate initial SOC1 (%)	Estimated Capacity1 (Ah)	Estimated capacity Error1 (%)	Initial SOC Error2 (%)	Inaccurate initial SOC2 (%)	Estimated Capacity2 (Ah)	Estimated capacity Error2 (%)
−5	5.54	2.560	0.170	5	15.54	2.578	0.891
	15.42	2.544	−0.423		25.42	2.565	0.376
	25.32	2.543	−0.486		35.32	2.565	0.366
	35.21	2.529	−1.036		45.21	2.553	−0.090
	45.23	2.521	−1.350		55.23	2.547	−0.314
	55.19	2.484	−2.778		65.19	2.516	−1.551

influence the work reported in this paper.

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